Continuum Mechanics
class # 19
10/22/2026

Local Lagrangian Form of the Second Law

Result (5.19): Let $\vec{\phi}: B \times [0,\infty) \to \mathbb{R}$ be a motion of a continuum body. Then the Second Law implies that

$$\rho_0 \dot{\eta}_m = \rho_0 \frac{R}{\Theta} - \nabla^X \cdot \left(\frac{\vec{Q}}{\Theta}\right) + \Gamma_m, \quad \forall \vec{X} \in B, \ \forall t \geq 0,$$

where

$$\Theta(\vec{X},t) = \theta_m(\vec{X},t) = \theta(\vec{\phi}(\vec{X},t),t),$$

$$R(\vec{X},t) = r_m(\vec{X},t) = r(\vec{\phi}(\vec{X},t),t),$$

and $$\vec{Q}(\vec{X},t) = \det(F(\vec{X},t)) F(\vec{X},t)^{-1} \vec{q}_m(\vec{X},t).$$

$$\Gamma_m \geq 0, \quad \forall \vec{X} \in B, \ \forall t \geq 0.$$

Proof: Axiom (5.4), the Second Law of Thermodynamics, states that

$$\frac{d}{dt}\int_{\Omega_t} \rho \eta \, dV_x = \int_{\Omega_t} \frac{\rho r}{\theta} dV_x - \int_{\partial\Omega_t} \frac{\vec{q}\cdot\vec{n}}{\theta} dA_x$$

for any $\Omega_t \subseteq B_t$. Changing variables, we have

$$\int_{\Omega_t} \rho \eta \, dV_x = \int_{\Omega_0} \rho_m \eta_m \det(F) dV_x$$

$$= \int_{\Omega_0} \rho_0(\vec{X}) \eta_m(\vec{X},t) \, dV_x,$$

where we have used the conservation of mass equation

$$\rho_m(\vec{x},t)\det(F(\vec{x},t)) = \rho_0(X), \quad \forall X \in B, t \geq 0.$$

Similarly,

$$\int_{\Omega_t} \frac{\rho r}{\theta}\, dV_x = \int_{\Omega_0} \frac{\rho_m r_m}{\theta_m} \det(F)\, dV_X$$

$$= \int_{\Omega_0} \rho_0(X)\, \frac{R(\vec{X},t)}{\Theta(\vec{X},t)}\, dV_X$$

Finally, using Result (4.11) and the Divergence Theorem,

$$\int_{\partial\Omega_t} \frac{\vec{q}\cdot\vec{n}}{\theta}\, dA_x = \int_{\partial\Omega_0} \frac{\vec{q}_m}{\theta_m} \cdot \left(\det(F)\, F^{-T}\vec{N}\right) dA_X$$

$$= \int_{\partial\Omega_0} \left(\frac{\vec{Q}}{\Theta}\right)\cdot\vec{N}\, dA_X$$

$$= \int_{\Omega_0} \nabla^X\cdot\left(\frac{\vec{Q}}{\Theta}\right) dV_X$$

Thus, the Second Law may be expressed as

$$\int_{\Omega_0} \rho_0 \dot{\eta}_m\, dV_X = \int_{\Omega_0} \left\{\rho_0 \frac{R}{\Theta} - \nabla^X\cdot\left(\frac{\vec{Q}}{\Theta}\right) + \sigma_m\right\} dV_X$$

Using the localization Theorem

$$\rho_0\dot{\eta}_m = \rho_0\frac{R}{\Theta} - \nabla^X\cdot\left(\frac{\vec{Q}}{\Theta}\right) + \sigma_m.$$

Furthermore,

$$\sigma_m \geq 0, \quad \forall \vec{X} \in \mathcal{B}, \ \forall t \geq 0. \qquad ///$$

Result (5.20): Let $\vec{\varphi} : \mathcal{B} \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body. Then, the Second Law of Thermodynamics implies that, for all $\vec{X} \in \mathcal{B}$, $t \geq 0$,

$$\rho_0 \dot{\Psi} = \mathbf{P} : \dot{\mathbf{F}} - \rho_0 \eta_m \dot{\Theta} - \frac{\vec{Q}}{\Theta} \cdot \nabla^X \Theta - \Theta \sigma_m$$

where

$$\Theta \sigma_m \geq 0, \quad \forall \vec{X} \in \mathcal{B}, \ \forall t \geq 0,$$

and

$$\Psi(\vec{X},t) := \mathcal{E}(\vec{X},t) - \Theta(\vec{X},t)\eta_m(\vec{X},t).$$

Proof: Exercise. ///

Frame Indifference

Frame indifference has to do with the fact that certain fields are independent of the observer.

Definition (5.21): Two motions $\vec{\varphi}, \vec{\varphi}^* : \mathcal{B} \times [0,\infty) \to \mathbb{E}^3$ of a body are related by a superposed rigid motion iff, for all $\vec{X} \in \mathcal{B}$, $t \geq 0$,

(5.36)
$$\vec{\varphi}^*(\vec{X},t) = \mathbf{Q}(t)\,\vec{\varphi}(\vec{X},t) + \vec{c}(t),$$

for some rotation $\mathbf{Q}(t)$ and vector $\vec{c}(t)$.

Recall $Q(t) \in \mathcal{V}^2$ is a rotation iff, for all $t \geq 0$,

$$Q(t)^T Q(t) = Q(t) Q(t)^T = I$$

and

$$\det(Q(t)) = 1. \quad //$$

We say that different pairs of observers are described by different functions $Q(t)$ and $\vec{c}(t)$.

We assume that different observers use the same synchronized clocks.

Let's put an example here. ↓ ↓

Result (5.22): Let $\vec{\varphi}, \vec{\varphi}^* : \mathcal{B} \times [0,\infty) \to \mathcal{E}^3$ be two motions related by (5.36). Then, the relation between material fields (F, R, U, V, C) and $(F^*, R^*, U^*, V^*, C^*)$ is given by

$$F^* = QF, \quad R^* = QR,$$

$$U^* = U, \quad V^* = QVQ^T, \quad C^* = C,$$

for all $\vec{X} \in \mathcal{B}$, $t \geq 0$. The relation between the spatial fields $(\nabla^x \vec{v}, L)$ and $(\nabla^{x^*} \vec{v}^*, L^*)$ is

$$\nabla^{x^*} \vec{v}^*(\vec{x}^*, t)\Big|_{\vec{x}^* = \vec{g}(\vec{x},t)} = Q(t) \nabla^x \vec{v}(\vec{x}, t) Q(t)^T + \dot{Q}(t) Q(t)^T$$

$$L^*(\vec{x}^*, t)\Big|_{\vec{x}^* = \vec{g}(\vec{x},t)} = Q(t)\, L(\vec{x}, t)\, Q(t)^T,$$

for all $\vec{x} \in B_t$, $t \geq 0$, where

$$\vec{g}(\vec{x},t) := Q(t)\vec{x} + \vec{c}(t). \quad ///$$

Proof: let us first recall some definitions:

$$F = VR = RU$$

are the left and right polar decompositions, where R is a rotation and

$$U = \sqrt{F^T F} \qquad V = \sqrt{F F^T}.$$

The Cauchy-Green Strain tensor is

$$C = F^T F.$$

Taking the $\vec{x}$-gradient of $\vec{\varphi}^*$ gives

$$\begin{aligned} F^*(\vec{x},t) &= \nabla^x \phi^*(\vec{x},t) \\ &= \nabla^x \left(Q(t)\vec{\phi}(\vec{x},t) + \vec{c}(t) \right) \\ &= \nabla^x \left(Q(t)\vec{\phi}(\vec{x},t) \right) \\ &= Q(t)\nabla^x \vec{\phi}(\vec{x},t) \\ &= Q(t) F(\vec{x},t). \end{aligned}$$

Now,

$$U^* = \sqrt{F^{*T} F^*} = \sqrt{F^T Q^T Q F} = \sqrt{F^T F} = U.$$

Now,

$$R^* U = R^* U^* = F^* = Q F = Q R U$$

Thus

$$R^* U = Q R U = F^*$$

are two polar decompositions for F^*. Note that both R^* and QR are rotation matrices. By uniqueness of the polar decomposition,

$$R = QR.$$

The rest of the details are left as exercises. ///

Definition (5.23): The fields ϕ, $\vec{w}$, and S are called frame-indifferent iff for all superposed rigid motions

$$\vec{x}^* = \vec{q}_t(\vec{x}) = \vec{g}(\vec{x}, t) = Q(t)\,\vec{x} + \vec{c}(t)$$

we have

$$\phi^*(\vec{x}^*, t) = \phi(\vec{x}, t),$$

$$\vec{w}^*(\vec{x}^*, t) = Q(t)\,\vec{w}(\vec{x}, t),$$

$$S^*(\vec{x}^*, t) = Q(t)\,S(\vec{x}, t)\,Q(t)^T,$$

for all $\vec{x} \in \mathcal{B}_t$ and $t \geq 0$.

Axiom (5.24) The spatial mass density, $\rho(\vec{x},t)$, temperature, $\theta(\vec{x},t)$, entropy per unit mass, $\eta(\vec{x},t)$, internal energy per unit mass, $\phi(\vec{x},t)$, Cauchy stress, $S(\vec{x},t)$, and Fourier heat flux, $\vec{q}(\vec{x},t)$, are all frame-indifferent fields. //

Proposition: Suppose that the internal energy, heat flux and Cauchy stress are modeled by

$$\phi(\vec{x},t) = \hat{\phi}\big(\rho(\vec{x},t), \theta(\vec{x},t), L(\vec{x},t)\big),$$

$$\vec{q}(x,t) = \hat{\vec{q}}\big(\rho(\vec{x},t), \theta(x,t), L(\vec{x},t)\big),$$

$$S(\vec{x},t) = \hat{S}\big(\rho(\vec{x},t), \theta(x,t), L(\vec{x},t)\big),$$

where $\hat{\phi}$, $\hat{\vec{q}}$, and $\hat{S}$ are functions. Then, if the axiom of frame indifference holds,

$$\hat{\phi}(\rho, \theta, QLQ^T) = \hat{\phi}(\rho, \theta, L),$$

$$\hat{\vec{q}}(\rho, \theta, QLQ^T) = Q\hat{\vec{q}}(\rho, \theta, L),$$

$$\hat{S}(\rho, \theta, QLQ^T) = Q\hat{S}(\rho, \theta, L)Q^T,$$

for all rotations Q and all admissible values of ρ, θ, and L.

Proof: By the Axiom of frame indifference and Defn (5.23),

$$\hat{\phi}(\rho^*, \theta^*, L^*) = \hat{\phi}(\rho, \theta, L)$$

$$\hat{\vec{q}}(\rho^*, \theta^*, L^*) = Q\hat{\vec{q}}(\rho, \theta, L)$$

and

$$\hat{S}(\rho^*, \theta^*, L^*) = Q\hat{S}(\rho, \theta, L)Q^T.$$

Since $\rho^* = \rho$ and $\theta^* = \theta$ by Axiom (5.24) and

$$L^* = QLQ^T$$

by Result (5.22), it follows that

$$\hat{\phi}(\rho, \theta, QLQ^T) = \hat{\phi}(\rho, \theta, L),$$

et cetera. ///

Thus, Frame Indifference imposes severe restrictions on the constitutive relations that are used to close the models. ///

Material Constraints

Definition (5.25): A continuous body is said to be subject to a simple material or internal constraint if every motion $\vec{\varphi}: \mathcal{B} \times [0, \infty) \to \mathbb{E}^3$ must satisfy an equation of the form

(5.37)
$$\gamma(F(\vec{X}, t)) = 0, \quad \forall X \in \mathcal{B}, \ t \geq 0,$$

where F is the deformation gradient and $\gamma: \mathcal{V}^2 \to \mathbb{R}$ is a given function.

Example: A material that admits no volume change is called incompressible. In view of Result (4.12) an appropriate constraint is

$$\gamma(F) = \det(F) - 1. \tag{5.38}$$

When $\gamma(F) = 0$, we have

$$\det(F) = 1, \quad \forall\, \vec{X} \in B, \; t \geq 0.$$

An equivalent form of the constraint is

$$\nabla^{x} \cdot \vec{v}(\vec{x}, t) = 0, \quad \forall\, \vec{x} \in B_t, \; t \geq 0,$$

in view of Result (4.12).

Axiom (5.26): The first Piola-Kirchhoff stress field $P(\vec{X}, t)$ in a continuum body subject to a material constraint

$$\gamma(F(\vec{X}, t)) = 0$$

can be decomposed into parts:

$$P(\vec{X}, t) = P^{(r)}(\vec{X}, t) + P^{(a)}(\vec{X}, t),$$

where $P^{(a)}$ is an active stress field determined by a constitutive equation and $P^{(r)}$ is a reactive stress field whose stress power is zero in all possible motions, that is

$$P^{(r)} \dot{F} = 0 \quad \forall\, \vec{X} \in B, \; t \geq 0. \tag{5.39}$$

Proposition: Suppose that Axiom (5.26) hold and

$$\gamma(F(\vec{X},t)) = 0$$

is an internal constraint for a material. Then the most general form of the reactive stress field is

$$P^{(r)}(\vec{X},t) = q(\vec{X},t)\, D\gamma(F(\vec{X},t)), \quad \forall \vec{X}\in B,\ t\geq 0,$$

where $q(\vec{X},t)$ is an unknown scalar field known as a multiplier. In particular, if the constraint is given by incompressibility, ie,

$$\gamma(F) = \det(F) - 1, \tag{5.38}$$

then

$$P^{(r)}(\vec{X},t) = q(\vec{X},t)\det(F(\vec{X},t))F(\vec{X},t)^{-T}. \tag{5.40}$$

Proof: From Equation (5.39)

$$P^{(r)} : \dot{F} = 0, \quad \forall \vec{X}\in B,\ t\geq 0.$$

Because of the constraint $\gamma(F)=0$, it follows that, for all $\vec{X}\in B$, $t\geq 0$,

$$0 = \frac{d}{dt}\gamma(F(\vec{X},t)) = D\gamma(F(\vec{X},t)) : \dot{F}(\vec{X},t)$$

So $\dot{F}$ is orthogonal to $D\gamma$. Thus, $P^{(r)}$ must be parallel to $D\gamma(F(\vec{X},t))$. Hence,

$$P^{(r)}(\vec{X},t) = q(\vec{X},t)\, D\gamma(F(\vec{X},t)),$$

where q is independent of F.

If $\gamma(F) = \det(F) - 1$

$$D\gamma(F) = \det(F)\, F^{-T}$$

by Result (2.22) and the proof is complete. ///

Proposition: Suppose that Axiom (5.26) hold and

$$\gamma(F) = 0$$

is a constraint on the system. Then the Cauchy stress field may be decomposed as

$$S(\vec{x},t) = S^{(r)}(\vec{x},t) + S^{(a)}(\vec{x},t)$$

where $S^{(r)}$ is a reactive Cauchy stress corresponding to $P^{(r)}$ and $S^{(a)}$ is an active Cauchy stress corresponding to $P^{(a)}$. In particular, in incompressible materials, the most general form of the reactive stress is

$$S^{(r)}(\vec{x},t) = -p(\vec{x},t)\, I$$

where

$$p(\vec{x},t) = q_s(\vec{x},t)$$

is the spatial description of the material field q and is usually called the pressure.

Proof: Exercise. ///

Isothermal Considerations

Many continuum processes are almost purely mechanical and can be approximated as isothermal.

Defn (5.27): By a *thermo-mechanical process* for a continuum body, we mean a pair of functions $(\vec{\phi}, \Theta)$, where $\vec{\phi}$ is an admissible motion and $\Theta(\vec{X}, t)$ is a material temperature field. A process is called *isothermal* if Θ is constant independent of $\vec{X}$ and t, equivalently if $\Theta(\vec{x}, t)$ is constant independent of $\vec{x}$ and t. //

Remark: Typically for isothermal processes we can neglect the balance of internal energy since this directly relates the heat flux, heat supply, and internal energy. The heat flux and heat supply are typically zero in isothermal models.

Ironically, perhaps, the Clausius-Duhem inequality is still important for isothermal modeling. ///

Definition (5.28): A thermo-mechanical process $(\vec{\phi}, \Theta)$ for a body is said to be closed in an interval $[t_0, t_1]$ if

$$\vec{\phi}(\vec{X}, t_0) = \vec{\phi}(\vec{X}, t_1), \quad \forall \vec{X} \in \mathcal{B},$$

$$\dot{\vec{\phi}}(\vec{X}, t_0) = \dot{\vec{\phi}}(\vec{X}, t_1), \quad \forall \vec{X} \in \mathcal{B},$$

and

$$\Theta(\vec{X}, t_0) = \Theta(\vec{X}, t_1), \quad \forall \vec{X} \in \mathcal{B}.$$

A body is called <u>energetically passive</u> if for every closed process its material free energy density satisfies

$$\underline{\Psi}(\vec{X}, t_1) \geq \underline{\Psi}(\vec{X}, t_0), \quad \forall \vec{X} \in \mathcal{B}.$$